Project EDA Report

Nocturnal VPR: Dataset and Experiment Diagnostics

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April 24, 2026

Purpose

This exploratory data analysis (EDA) was designed to answer a simple question before adding more model complexity: *what is actually difficult about our data, and what do the current results suggest about the next useful step?*

The analysis combines:

- image statistics for **GardensPoint** and the **strict campus dataset**,
- label and place-coverage analysis for the campus data, and
- a comparison of descriptor results already saved in the experiment run folders.

How The EDA Was Run

The code for this EDA is in `analysis/project_eda.py`. It loads the same datasets already used by the benchmark scripts, computes per-image statistics, parses saved experiment result files, and writes plots into `output_images/runs/eda/`.

Question 1: How strong is the day-night appearance gap?

Answer: The answer depends on the dataset, and that difference matters.

Split	Brightness Mean	Contrast Mean	Sharpness Mean
GardensPoint day	97.1	58.9	405.8
GardensPoint night	127.6	69.8	718.3
Campus day	112.0	53.5	784.2
Campus night	79.1	48.4	719.7

Interpretation.

- On **GardensPoint**, the night images are on average **brighter**, more contrasty, and sharper than the day images.
- On **campus**, the night images are clearly **darker** and slightly lower in contrast than the day images.

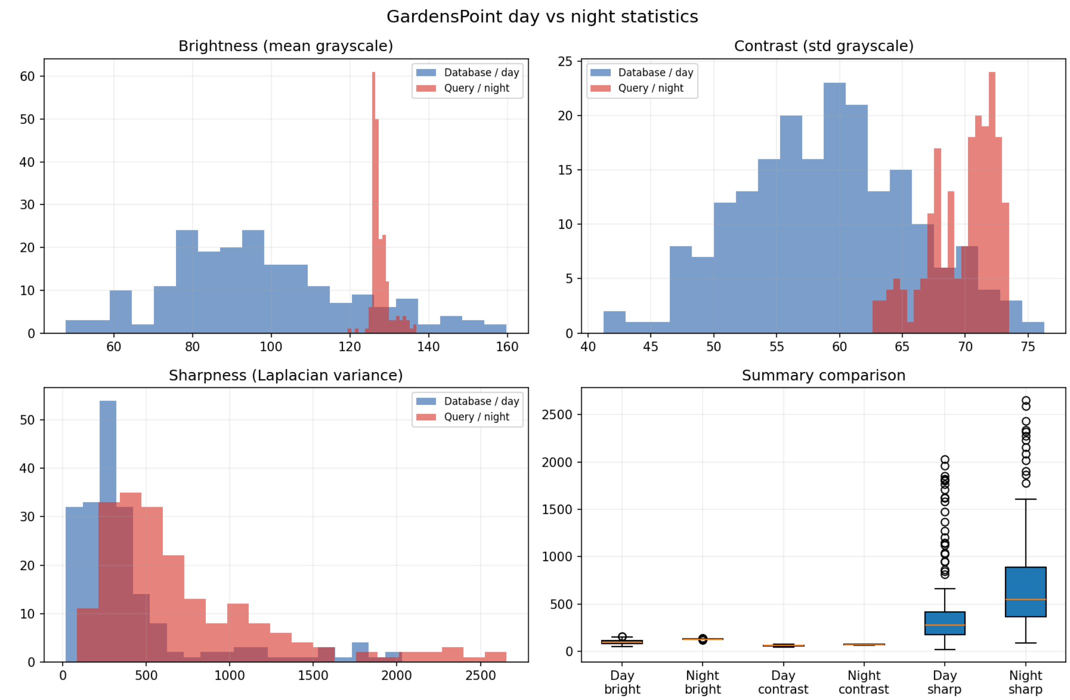

Figure 1: GardensPoint day versus night image statistics.

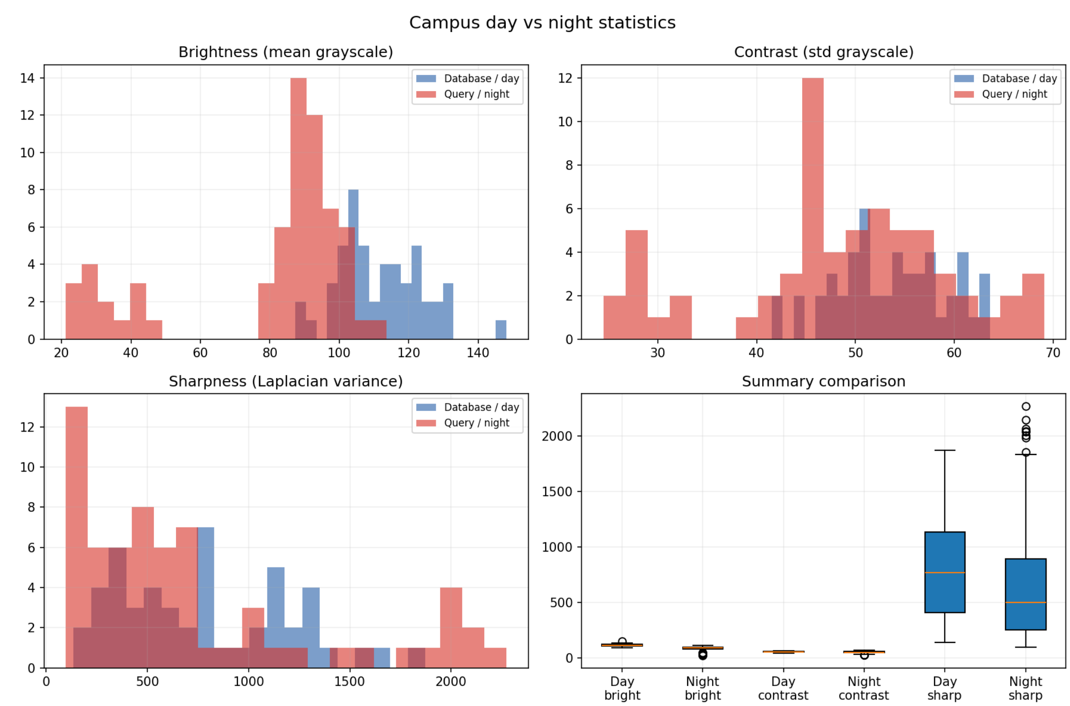

Figure 2: Campus day versus night image statistics.

- Visually, the GardensPoint night frames appear exposure-compensated or over-brightened, which explains why their average intensity is unexpectedly high.

- This means the two datasets do not represent the same kind of night-time difficulty. GardensPoint is still useful as a controlled benchmark, but it does not fully reflect the darker and noisier campus setting.

Question 2: Is the strict campus protocol inherently hard?

Answer: Yes. The strict campus setup is hard not only because of descriptor quality, but also because many queries are treated as no-match cases.

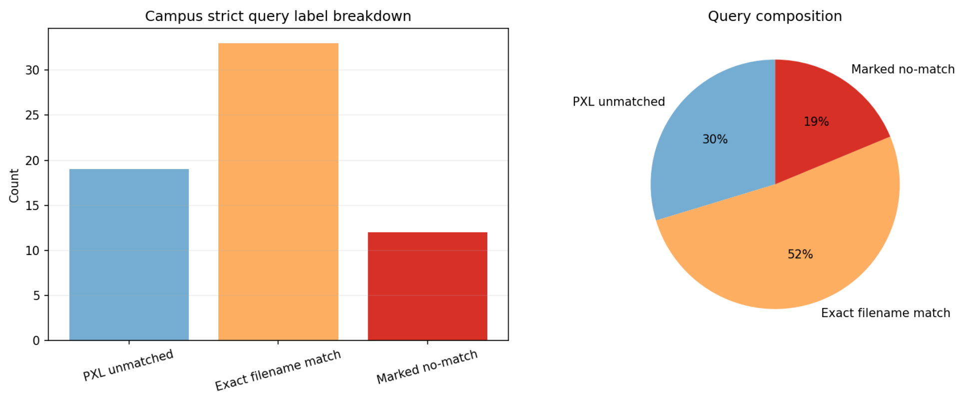

Figure 3: Strict campus query composition.

The strict campus query set contains:

- **33** exact filename matches,
- **12** queries explicitly marked as no-perfect-match, and
- **19** `PXL_*` phone images with no strict day counterpart.

So nearly half of the strict campus query set is not a simple one-to-one retrieval problem. This helps explain why thresholded precision-style metrics remain low even when top-k retrieval is strong. These `PXL_*` images are intentionally kept because they act as hard no-match queries in the strict setting.

Question 3: Is place coverage balanced across the campus walk?

Answer: Not really. The campus place-level annotations reveal an uneven distribution of visual evidence.

Main observations.

- The largest night bucket is `entrance` with **25** night images.
- Some places have very little support, such as `ncafe` and `cafbanner`.
- `hallway` has day images but no confident night counterpart in the place-level rematch.

This uneven coverage means that some locations are much more likely to dominate the matching statistics than others. When a descriptor performs well, part of that success may come from

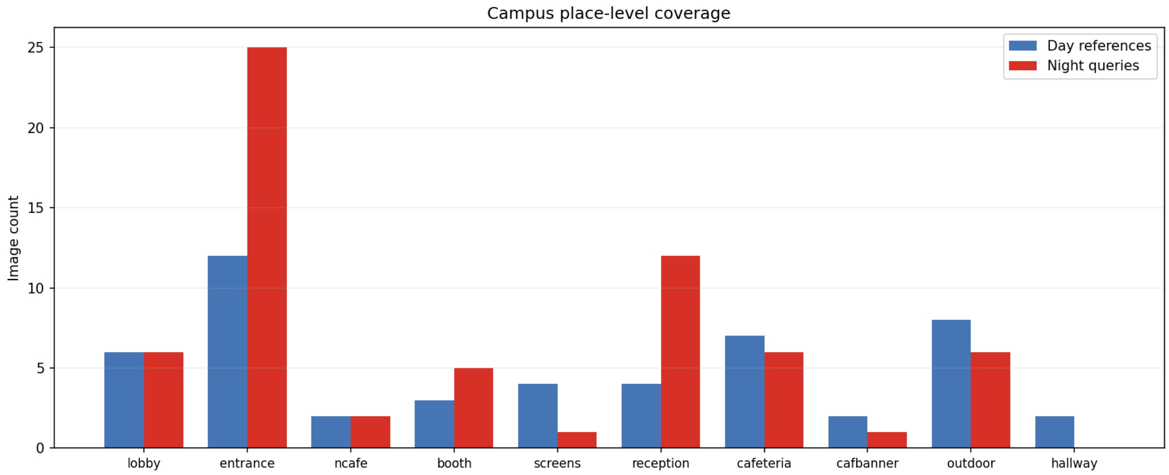

Figure 4: Place-level day and night coverage across the campus walk.

repeatedly seeing the same overrepresented place types rather than solving every place equally well.

Question 4: Which descriptors currently look strongest?

Answer: The current evidence supports modern VPR descriptors, but their relative strengths differ by metric.

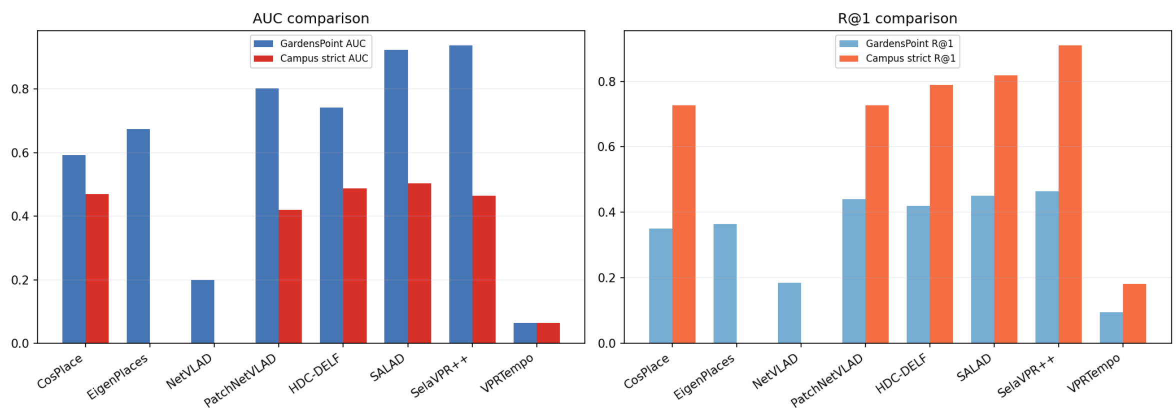

Figure 5: AUC and R@1 comparison across the saved GardensPoint and strict campus runs.

From the saved runs:

- **GardensPoint AUC**: best is **SelaVPR++** at **0.937**.
- **GardensPoint R@1**: best is **SelaVPR++** at **0.465**.
- **Strict campus AUC**: best is **SALAD** at **0.504**.
- **Strict campus R@1**: best is **SelaVPR++** at **0.909**.

Interpretation.

- **SALAD** and **SelaVPR++** are the most promising current descriptors.
- **CosPlace** remains a strong and simpler baseline.
- **VPRTempo** transfers poorly in this setup, which is consistent with the fact that the checkpoint we tested is a specialized demo model rather than a universal descriptor.

Question 5: What does this imply for the project?

Answer: The EDA supports the current project direction.

1. **Descriptor exploration is justified.** The descriptor choice clearly changes performance in a meaningful way.
2. **GardensPoint is still worth keeping.** It remains the right controlled day-to-night benchmark, even though it is not identical to campus.
3. **The campus challenge is broader than descriptor weakness alone.** It includes harder no-match composition, small-data imbalance, and viewpoint ambiguity.
4. **The next useful step is not random model complexity.** The most sensible next phase is to keep the strong descriptors and add lightweight improvements for rejection and temporal stability, such as CLAHE, sequence matching, or other temporal cues.

Conclusion

The EDA clarifies an important point: **our benchmark and deployment datasets represent different kinds of night-time difficulty**. GardensPoint tests controlled day-to-night retrieval, while the strict campus set adds darker imagery, harder no-match cases, and uneven place coverage. This makes the current results easier to trust and easier to interpret. The strongest descriptors so far are SALAD and SelaVPR++, and the remaining gap is increasingly about deployment realism rather than only about retrieval quality.